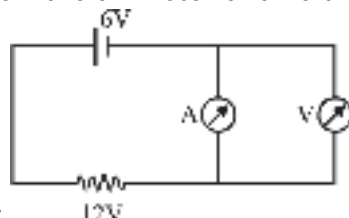


PHYSICS

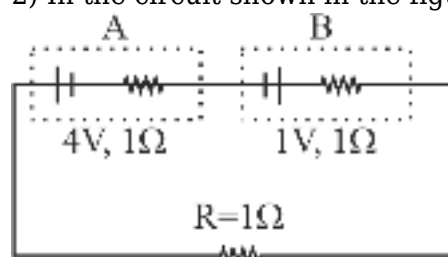
1) In the circuit given below the ammeter and voltmeter are ideal measuring devices. What is the



reading of the voltmeter:-

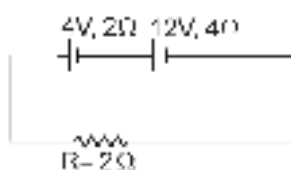
- (1) 6 V
- (2) 3 V
- (3) 1.5 V
- (4) zero

2) In the circuit shown in the figure, match the following two columns :



	Column I		Column II
(a)	Potential difference across battery A	(p)	zero
(b)	Potential difference across battery B	(q)	1
(c)	Net power consumed by R	(r)	2
(d)	Potential difference across R	(s)	3

- (1) a-p; b-q; c-r; d-s
- (2) a-s; b-q; c-r; d-p
- (3) a-s; b-r; c-q; d-q
- (4) a-q; b-r; c-s; d-s



3)

Current passing through resistance R will be :-

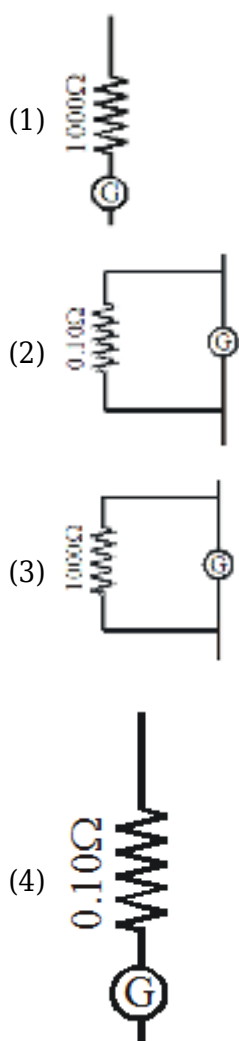
- (1) 2Amp
- (2) 1Amp

- (3) 4Amp
- (4) 0.5Amp

4) On changing the position of battery and galvanometer in wheat stone bridge respectively the new balancing stage :-

- (1) Remains unchangeable
- (2) Will change
- (3) It may change and not; depends on the resistances of battery and galvanometer
- (4) None can be said

5) A galvanometer with an internal resistance of 100Ω will show a full scale deflection with a current of 10 mA. Which of the following circuits would convert this galvanometer into an ammeter which will read 10A at full scale?

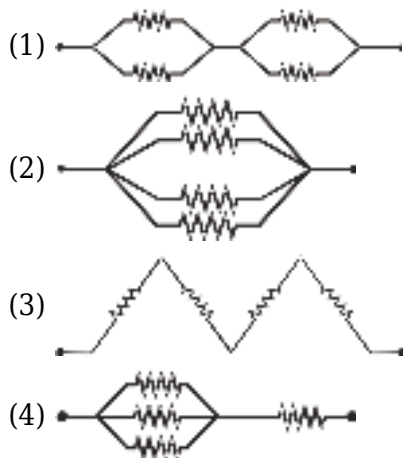


6) A galvanometer of resistance 20 ohm suffers full scale deflection for a current of 20 mA. It can be converted into a voltmeter of range 200 V by :-

- (1) connecting in series a resistance of 9980 Ω
- (2) connecting in series a resistance of 19980 Ω
- (3) using a shunt of resistance 20 m Ω

(4) using a shunt of resistance $9980 \text{ m}\Omega$

7) Which arrangement of four identical resistances should be used to draw maximum energy from a cell of voltage V :-

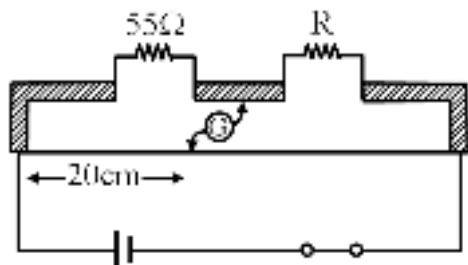


8) The potential difference ($V_A - V_B$) between the points A and B in the given figure is :-



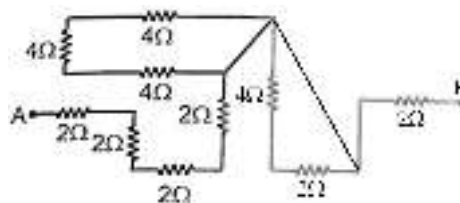
- (1) + 6 V
- (2) + 9 V
- (3) - 3 V
- (4) + 3 V

9) Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer.



The value of the unknown resistor R is

- (1) 220Ω
- (2) 110Ω
- (3) 55Ω
- (4) 13.75Ω



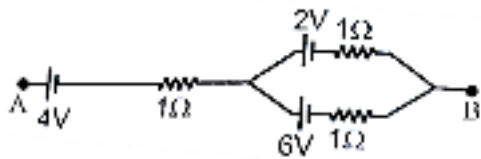
10) The equivalent resistance between A and B is :

- (1) 8Ω
- (2) 12Ω

(3) $10\ \Omega$

(4) $4\ \Omega$

11) Find the net emf of the three batteries shown in figure between A and B.



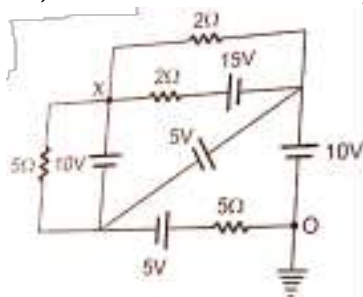
(1) 8 V

(2) 6 V

(3) 2 V

(4) 3 V

12) In the circuit shown if point O is earthed, the potential of point X is equal to :



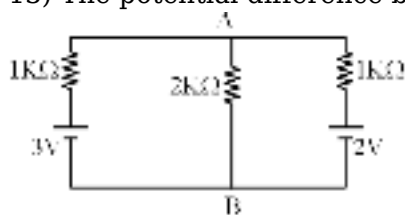
(1) 10V

(2) 15 V

(3) 25 V

(4) 12.5 V

13) The potential difference between points A and B in the given diagram is :



(1) 1 V

(2) 0.5 V

(3) 4 V

(4) 2 V

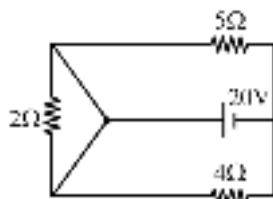
14) If a metal wire radius r is decreased by stretching it then its resistivity :

(1) constant

(2) $\propto \frac{1}{r^2}$

(3) $\propto \frac{1}{r^4}$

(4) $\propto \frac{1}{r^3}$

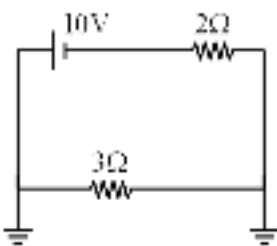


15) In the circuit shown in the figure, is zero

- (a) current passing through 2Ω resistance
 (b) current passing through 4Ω resistance is 5 A
 (c) current passing through 5Ω resistance is 4 A
 (d) potential difference across 2Ω resistance is zero
 which of following option will be correct

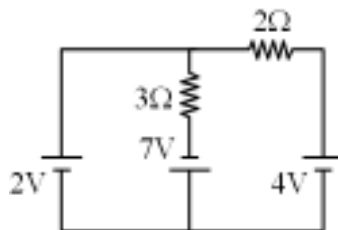
(a) current passing through 2Ω resistance

- (1) only a
 (2) a and c
 (3) a and d
 (4) all a, b, c, d



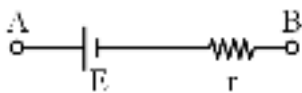
16) What is the current given by battery

- (1) Zero
 (2) 5 A
 (3) 2 A
 (4) 1 A



17) What will be the current in 3Ω resistance

- (1) 3 A
 (2) 4 A
 (3) 1 A
 (4) Zero



18)

In given part of circuit potential of A is higher than B then direction of current

- (a) may be from A to B
- (b) must be from A to B
- (c) may be from B to A
- (d) current may be zero

which of the following will be correct

- (1) only b
- (2) a and c
- (3) a, c, d
- (4) all a, b, c, d

19) If the vectors $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$ and $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$ are perpendicular to each other then the positive value of a is :-

- (1) Zero
- (2) 1
- (3) 2
- (4) 3

20) A vector having magnitude 30 unit makes equal angle with each of x, y and z axes. The components of vector along each of x, y and z axes are :-

- (1) $10\sqrt{2}$ units
- (2) $30\sqrt{2}$ units
- (3) $10\sqrt{3}$ units
- (4) $30\sqrt{3}$ units

21) Two forces of magnitude 8N and 15 N respectively act at a point. If the resultant force is 17 N. The angle between the forces has to be :-

- (1) 60°
- (2) 45°
- (3) 90°
- (4) 30°

22) If $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + 2\hat{k}$ find the magnitude of component of $(\vec{A} + \vec{B})$ along $\vec{B}$:-

- (1) 4 unit
- (2) 5 unit
- (3) 6 unit
- (4) 7 unit

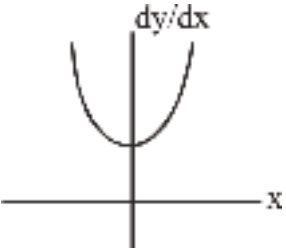
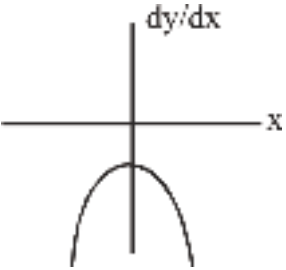
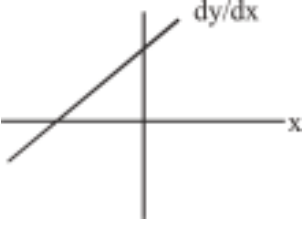
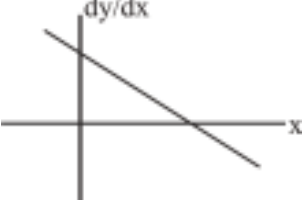
23) $\vec{A}$ and $\vec{B}$ are two vectors and θ is the angle between them. If $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$ the value of θ is :-

- (1) 45°
- (2) 30°
- (3) 90°
- (4) 60°

24) The magnitudes of x and y components of $\vec{P}$ are 7 and 6. Also the magnitudes of x and y components $\vec{P} + \vec{Q}$ are 11 and 9 respectively. What is the magnitude of Q :-

- (1) 5
- (2) 6
- (3) 8
- (4) 9

25) If $y = 4x^2 + 2x$ then graph between $\frac{dy}{dx}$ and x is :-

- (1) 
- (2) 
- (3) 
- (4) 

26) A particle is moving in a straight line according to the following velocity-time relation.

$$V = 2t + 3 \text{ m/s}$$

Find out velocity of the particle up to 1 sec. in m/s.

- (1) 8

- (2) 3
(3) 5
(4) 4

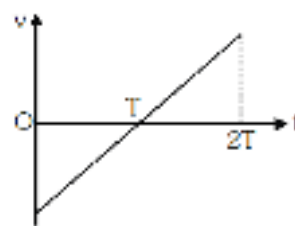
27) A ball thrown upwards covers distance in first second which is three times the distance in the last 3 seconds of ascent. What is initial velocity of projection?

- (1) 120 m/s
(2) 130 m/s
(3) 160 m/s
(4) 140 m/s

28) A particle moving along a straight line with uniform acceleration has velocities 7m/s at A and 17m/s at C. B is the mid point of AC. Then the incorrect option is :

- (1) The velocity at B is 13 m/s
(2) The average velocity between A and B is 10m/s
(3) The ratio of the time to go from A to B to that from B to C is 3 : 2
(4) The average velocity between B and C is 14 m/s

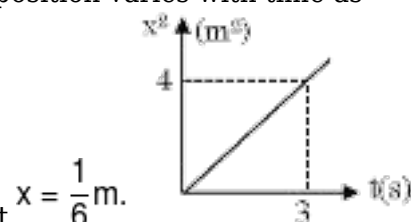
29) For a particle moving rectilinearly, the velocity-time ($\vec{v}-t$) graph is plotted. Which of the



following argument is incorrect to explain the facts about its motion ?

- (1) The acceleration of the particle remains constant.
(2) The particle changes its direction of motion at some point.
(3) The initial and final speeds of the particle are the same.
(4) The displacement of the particle is not zero

30) A particle is moving along a straight line such that square of its position varies with time as

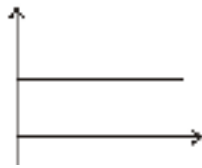
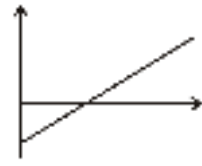
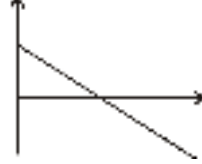
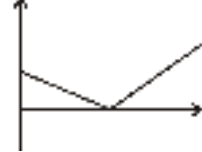
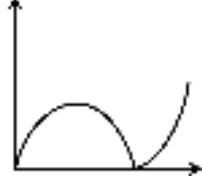


shown in the following graph. What is the velocity (m/s) of particle at $x = \frac{1}{6}$ m.

- (1) 4/3
(2) 2/9
(3) 9/2
(4) 4

31) A particle is moving along x-axis, such that its position varies with time as $x = at^2 + bt$, ($a > 0$, b

< 0). List-I contains the parameters on y and x-axis respectively and List-II contains some graphs.

List-I		List-II	
(P)	Acceleration-time	(1)	
(Q)	Distance-time	(2)	
(R)	Velocity-time	(3)	
(S)	Speed time	(4)	
		(5)	
		(6)	

(1) P-4, Q-1, R-3, S-2

(2) P-1, Q-6, R-3, S-5

(3) P-1, Q-2, R-3, S-5

(4) P-5, Q-6, R-1, S-4

32) A body starts from the origin and moves along the X-axis such that the velocity at any instant is given by $(4t^3 - 2t)$, where t is in second and velocity in m/s. What is acceleration (in m/s^2) of the particle, when it is 2 m from the origin.

(1) 0

(2) 4

(3) 16

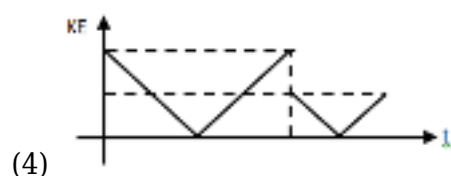
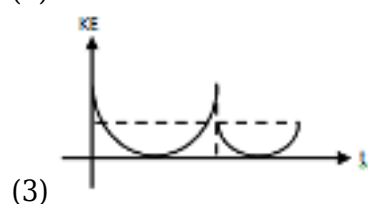
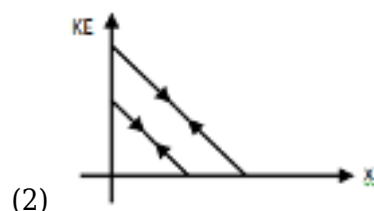
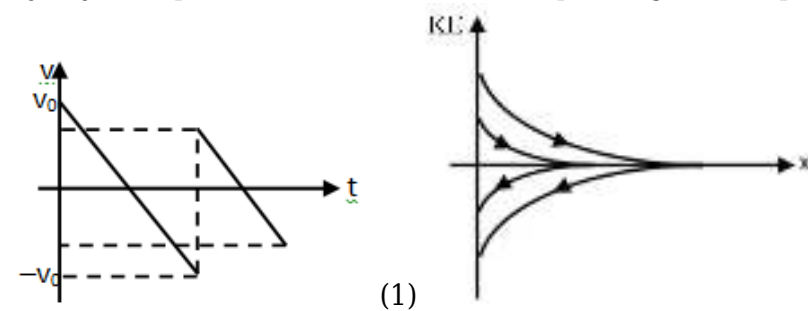
(4) 22

33) If particle starts from rest and assume constant acceleration. If ratio of distance covered in first

one second and first two seconds is $\frac{p}{q}$ then what is the value of $p + q$. (where p and q are minimum integers)

- (1) 5
- (2) 8
- (3) 11
- (4) 3

34) A particle is moving in straight line along x-axis such that velocity (v) depends on time (t) given by adjacent plot. Then choose the corresponding correct plot (KE is kinetic energy)



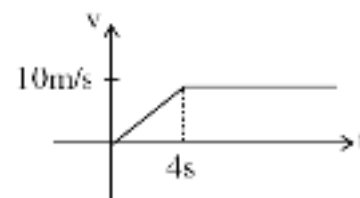
- (1) Only 2
- (2) Only 4
- (3) Both (1) and (4)
- (4) Both (2) and (3)

35) A ball is projected vertically upward from the top of a tower such that it crosses a point after 3 sec and 5 sec of projection. Ball will strike the ground in 2 sec if it is projected vertically downward with same speed, from tower. Then

List-I		List-II	
(P)	$\frac{\text{Average speed}}{ \text{Average velocity} }$	(1)	8

(Q)	$\frac{ \vec{a}_{avg} }{g}$	(2)	5
(R)	$\frac{\text{Initial speed}}{v}$ v - average speed from 3 to 5 sec	(3)	2.6
(S)	$\frac{ \text{change in velocity} }{ \text{initial velocity} }$	(4)	1
		(5)	2.5

- (1) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 5$
 (2) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$
 (3) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$
 (4) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$



36) Time taken to reach from starting point to displacement of 120 m is :

- (1) 8 s
 (2) 10 s
 (3) 12 s
 (4) 14 s

37) **Statement-I:** In free fall, the rate of change of velocity with time is constant for all objects.

Statement-II: In free fall, the change in velocity with distance increases with increasing distance of fall.

- (1) Both **Statement-I** and **Statement-II** are true.
 (2) Both **Statement-I** and **Statement-II** are false.
 (3) **Statement-I** is false and **Statement-II** is true.
 (4) **Statement-I** is true and **Statement-II** is false.

38) A car starting from rest, accelerate uniformly at the rate of 10 m/s^2 & retards uniformly at the rate of 20 m/s^2 to come to rest. Find out average velocity to complete the journey of 3 km.

- (1) 25 m/s
 (2) 50 m/s
 (3) 100 m/s
 (4) None

Common Content for Question No. 39 to 40

Two balls A and B are thrown with same speed u from the top of a tower. Ball A is thrown vertically upwards and the ball B is thrown vertically downwards.

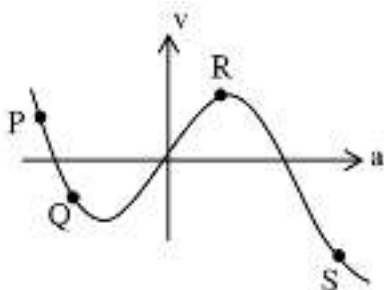
39) If t_A and t_B are the respective times taken by the balls A and B respectively to reach the ground. If $t_A = 6$ sec and $t_B = 2$ sec, then find out height of the tower.

- (1) 80 m
- (2) 60 m
- (3) 45 m
- (4) 30 m

40) The initial speed of each ball is :

- (1) 10 m/s
- (2) 15 m/s
- (3) 20 m/s
- (4) 40 m/s

41) Acceleration-velocity graph of a moving particle is shown in figure. The particle is

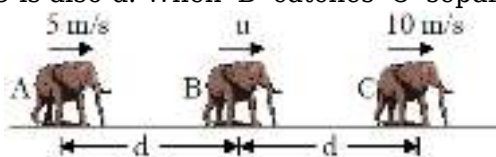


- (1) speeding up at P
- (2) speeding up at Q
- (3) speeding up at S
- (4) speeding down at R

42) If position of a particle is given by $x = 8t - t^2 + 2$, where x is in meters & t in second, then distance travelled by particle during $t = 0$ to $t = 6$ sec will be

- (1) 12 m
- (2) 28 m
- (3) 20 m
- (4) 0 m

43) Three elephants A, B and C are moving along a straight line with constant speed in same direction as shown in figure. Speed of A is 5 m/s and speed of C is 10 m/s. Initially separation between A and B is 'd' and between B and C is also d. When 'B' catches 'C' separation between A and



C becomes $3d$. Then the speed of B will be

- (1) 15 m/s
- (2) 7.5 m/s

(3) 20 m/s

(4) 5 m/s

44) A lift is moving downward with constant velocity 10 m/s. A coin is dropped at height 5 m from the floor of lift. Find distance travelled by coin in ground frame to reach the floor :-

(1) 10 m

(2) 5 m

(3) 20 m

(4) 15 m

45) On a two-lane road, car A is travelling with a speed of 36 km h^{-1} . Two cars B and C approach car A in opposite directions with a speed of 54 km h^{-1} each. At a certain instant, when the distance AB is equal to AC both being 1 km. B decides to overtake A before C does. What minimum acceleration of car B is required to avoid an accident ?

(1) 2 m/s^2

(2) 4 m/s^2

(3) 1 m/s^2

(4) 0.5 m/s^2

CHEMISTRY

1) For a reaction $A \rightarrow P$; A plot between $[A]^0$ Vs $\frac{1}{t_{1/2}}$ is a straight line having a positive slope when the initial concentration of $1 \times 10^{-2} \text{ M}$ its half-life period is found to be 20 min. When the concentration of A is $2 \times 10^{-2} \text{ M}$. Then half-life will be ?

(1) 20 min.

(2) 10 min.

(3) 67 min.

(4) 56 min.

2) At 27°C , for the reaction,

$\text{N}_2\text{O}_{4(g)} \rightarrow 2\text{NO}_{2(g)}$, rate of reaction is $6 \times 10^{-3} \text{ atm min}^{-1}$. Find rate of same reaction in terms of $\text{mol L}^{-1} \text{ min}^{-1}$.

(1) 1.2×10^{-4}

(2) 2.4×10^{-4}

(3) 3.2×10^{-2}

(4) 6×10^{-3}

3) 1st order reaction $A \rightarrow B + C$, carried out at 27°C if $3.8 \times 10^{-16} \%$ of the reactant molecules exist in the activated state, the E_a of the reaction is :-

(1) 12 KJ/mol

- (2) 831.4 KJ/mol
(3) 100 KJ/mol
(4) 88.57 KJ/mol

4) For reaction $A \rightarrow B$ concentration is expressed as $C_t = C^0 e^{-kt}$ calculate rate of reaction at $t = \frac{1}{k}$ sec.
If $C_0 = 10 \frac{\text{mol}}{\text{lit.}}$ & $k = 500 \text{ sec}^{-1}$

- (1) 5000
(2) $\frac{5000}{e}$
(3) 5000 e
(4) $\frac{e}{5000}$

5) A solution of N_2O_5 in CCl_4 yields 5 mL of O_2 by decomposition after 20 minutes from the start of the experiment and 10 mL of O_2 after a very long time. The decomposition obeys first order kinetics. What volume of O_2 would have evolved, 40 minutes after the start

- (1) 7.5 mL
(2) 7.25 mL
(3) 1.5 mL
(4) 6 mL

6) **Assertion (A) :-** All collisions of reactant molecules lead to formation of product.

Reason (R) :- Only those collisions in which molecules have correct orientation and sufficient energy lead to formation of product.

- (1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct, but R is not the correct explanation of A.
(3) A is correct but, R is incorrect.
(4) R is correct but, A is incorrect.

7) In the following reaction, $2H_2O_2 \rightarrow 2H_2O + O_2$

rate of formation of O_2 is $36 \text{ g L}^{-1} \text{ min}^{-1}$, what is rate of disappearance of H_2O_2 in $\text{mol L}^{-1} \text{ min}^{-1}$

- (1) 2.25
(2) 3.50
(3) 1.50
(4) 2.75

8) If 'a' is the initial concentration of reactant, then the half life period of a reaction of n^{th} order is directly proportional to -

- (1) a^n
(2) a^{n-1}

(3) a^{1-n}

(4) a^{n+1}

9) Consider the reaction $2A + B \rightarrow \text{Products}$.

When concentration of B alone was doubled the half life did not change. When the concentration of A alone was doubled, the rate increased by two times. The unit of rate constant for this reaction is :

(1) $\text{L mol}^{-1} \text{sec}^{-1}$

(2) $\text{mol L}^{-1} \text{sec}^{-1}$

(3) s^{-1}

(4) Unit less

10) The plot of $\ln k$ versus $1/T$ is linear with slope of :

(1) $-\frac{E_a}{R}$

(2) $\frac{E_a}{R}$

(3) $\frac{E_a}{2.303R}$

(4) $-\frac{E_a}{2.303R}$

11) Which of the following statements are correct :

(a) Time required for 99.9% completion is 5 times of time required for 75% completion for a first order reaction.

(b) Time required for 99% completion is twice the time required for 90% completion for a first order reaction.

(c) Time required for 87.5% completion is 3 times of half life for zero order reaction.

(d) Time required for 80% completion is 4 times of time required for 20% completion for zero order reaction.

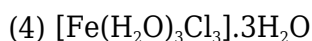
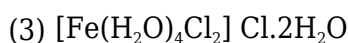
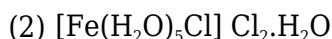
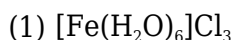
(1) (a) & (b) only

(2) (b) & (c) only

(3) (a), (b) & (c)

(4) (a), (b) & (d)

12) For 1 molal solution of each compound maximum freezing point will be assuming complete ionisation in each case :-



13) Which of the following is less than zero for ideal solution :-

(1) ΔH_{mix}

(2) ΔV_{mix}

(3) ΔG_{mix}

(4) ΔS_{mix}

14) The vapour pressure of a solvent decreased by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2, what would be the mole fraction of solvent if the decrease in vapour pressure is 20 mm of Hg. :-

(1) 0.8

(2) 0.6

(3) 0.4

(4) 0.2

15) Which of the following aqueous solution will have the lowest freezing point ?

(1) 0.10 m-sucrose

(2) 0.10 m-NiCl₂

(3) 0.10 m-CuSO₄

(4) 0.10 m-NH₄NO₃

16) In a 0.2 molal aqueous solution of a weak acid HX, the degree of ionization is 0.3. The freezing point of the solution will be nearest to (if k_f for H₂O is 1.86°C kg mol⁻¹)

(1) - 0.360 °C

(2) - 0.260 °C

(3) + 0.480 °C

(4) - 0.480 °C

17) Glucose is added to 1 litre water to such an extent that $\frac{\Delta T_f}{K_f}$ becomes equal to $\frac{1}{1000}$, the wt. of glucose added is :

(1) 180 g

(2) 18 g

(3) 1.8 g

(4) 0.18 g

18) **Assertion** : 0.1 m aqueous solution of glucose has higher elevation in the boiling point than 0.1 m aqueous solution of urea.

Reason : K_b of water for both 0.1 m aqueous solution of glucose and 0.1 m aqueous solution of urea has different values.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Both assertion and reason are incorrect.

19) What is the osmotic pressure of the solution obtained by mixing 300 cm³ of 0.6% w/v solution of urea with 300 cm³ 3.42% w/v solution of sucrose at 20°C ?

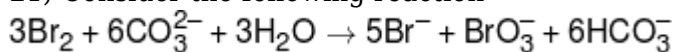
- (1) 5 atm
- (2) 5.2 atm
- (3) 2.4 atm
- (4) 4.5 atm

20) **Statement-I :-** Azeotrope is a binary mixture formed by ideal solutions.

Statement-II :- Azeotrope boils constant temperature.

- (1) Both statement-I and statement-II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Both statement-I and II are incorrect
- (4) Statement-I is incorrect but II is correct

21) Consider the following reaction



Which of the following statements is true regarding this reaction?

- (1) Bromine is oxidized and the carbonate is reduced
- (2) Bromine is reduced and the carbonate is oxidized.
- (3) Bromine is neither reduced nor oxidized.
- (4) Bromine is both reduced and oxidized

22) Match the column :-

	Column - A		Column - B (O.N.)
p	NH_3	(i)	+1
q	O_2F_2	(ii)	-3
r	CH_3COOH	(iii)	-1
s	BaO_2	(iv)	+3

- (1) p - (i), q - (ii), r - (iii), s - (iv)
- (2) p - (ii), q - (iii), r - (iv), s - (i)
- (3) p - (ii), q - (i), r - (iv), s - (iii)
- (4) p - (iii), q - (ii), r - (i), s - (iv)

23) On the basis of structure the oxidation state of two Cl atoms in CaOCl_2 respectively are :-

- (1) -1 and +1
- (2) +2 and -2
- (3) -2 and +2
- (4) -1 and +3

24) $\text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \rightarrow \text{Cr}^{3+} + \text{H}_2\text{O} + x\text{e}^-$ value of x is :-

- (1) 3
- (2) 6
- (3) 12
- (4) 8

25) $\text{Zn} + \text{NO}_3^- \rightarrow \text{Zn}^{+2} + \text{NH}_4^+$

The coefficient of NO_3^- , Zn and H^+ in balanced equation respectively are :-

- (1) 4, 1, 10
- (2) 10, 4, 1
- (3) 1, 4, 10
- (4) 4, 1, 7

26) In the reaction : $\text{FeS}_2 + \text{KMnO}_4 + \text{H}^+ \rightarrow \text{Fe}^{3+} + \text{SO}_2 + \text{Mn}^{2+} + \text{H}_2\text{O}$ the equivalent mass of FeS_2 would be equal to:

- (1) $\frac{M}{11}$
- (2) $\frac{M}{13}$
- (3) M
- (4) $\frac{M}{10}$

27) Which of the following is incorrect oxidation state ?

- (1) $\overset{*}{\text{N}}\text{H}_4\overset{*}{\text{N}}\text{O}_3 \rightarrow +1, -1$
- (2) $\text{Ca}(\overset{*}{\text{O}}\overset{*}{\text{C}}\overset{*}{\text{I}})\overset{*}{\text{C}}\overset{*}{\text{I}} \rightarrow +1, -1$
- (3) $\overset{*}{\text{Fe}}_3\overset{*}{\text{O}}_4 \rightarrow +3, +3, +2$
- (4) $\overset{*}{\text{Pb}}_3\overset{*}{\text{O}}_4 \rightarrow +4, +2, +2$

28) In a solution having solute (molar mass = m_B); density of Aq. solution is $d_{\text{g/ml}}$

Column-I		Column-II	
P	x% w/w	A	molarity = $\frac{x}{m_B} \times \frac{d}{100} \times 1000$
Q	x% w/v	B	molality = $\frac{M}{(1000d - Mm_B)} \times 1000$
R	M molar	C	molarity = $\frac{m \times d}{(1000 + mm_B)} \times 1000$

S	m molal	D	$\text{molarity} = \frac{x \times 10}{m_B}$
T	$\square_{\text{solute}}$	E	$\frac{\text{molality} \times 18}{1000}$

The correct match of column-I with column-II is-

- (1) (P-B), (Q-A), (R-C), (S-D), (T-E)
- (2) (P-A), (Q-D), (R-B), (S-C), (T-E)
- (3) (P-C), (Q-B), (R-A), (S-E), (T-D)
- (4) (P-B), (Q-A), (R-D), (S-C), (T-E)

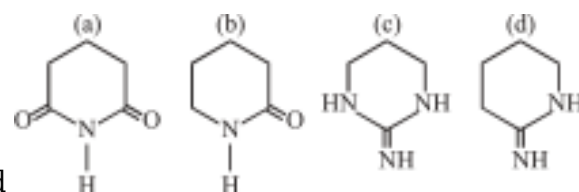
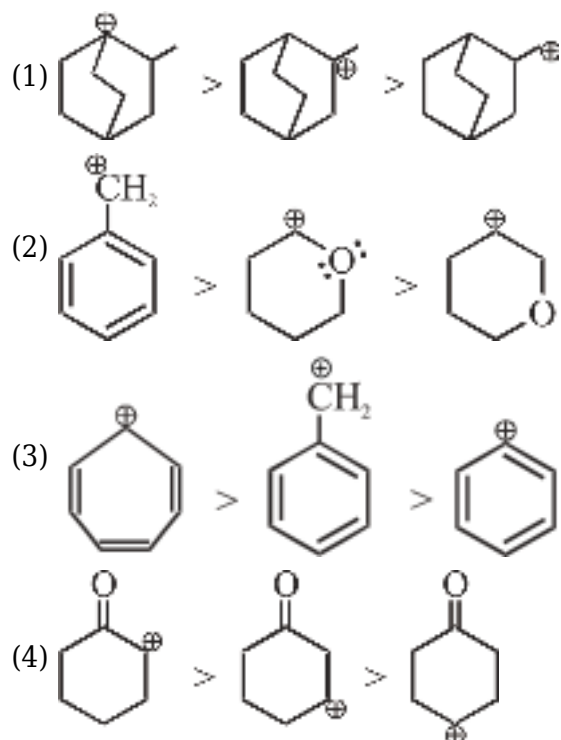
29) Find out correct order

- (1) $-\text{CH}_3 < -\text{CH}_2-\text{CH}_3 < -\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{CH}}} < -\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_3$ (+H order)
- (2) $-\overset{+}{\text{CH}_2} > -\text{NO}_2 > -\text{C}\equiv\text{N} > -\overset{\text{O}}{\parallel}{\text{C}}-\text{H} > -\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (-M order)
- (3) $-\overset{\ominus}{\text{CH}_2} > -\overset{\ominus}{\text{O}} > -\ddot{\text{N}}\text{H}_2 > -\ddot{\text{O}}\text{R} > -\ddot{\text{O}}\text{H}$
 $-\ddot{\text{N}}\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (+M order)
- (4) $-\overset{\ominus}{\text{O}} > -\overset{\text{O}}{\parallel}{\text{C}}-\overset{\ominus}{\text{O}} > -\text{CH}_3 > -\text{CD}_3 > -\text{CT}_3$ (+I order)

30) Which is correct basicity order of following compounds :-

- (1)
- (2)
- (3)
- (4)

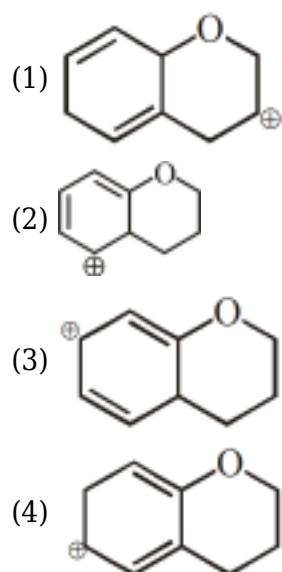
31) Which is correct stability order of following carbocation :-



32) Compare the Basic strength in the given compound

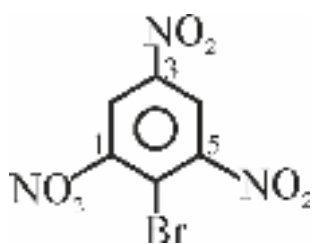
- (1) $a > b > c > d$
- (2) $c > d > a > b$
- (3) $c > d > b > a$
- (4) $d > c > b > a$

33) Which of the following is most stable cation?



34)

Select the correct statement about this compound.

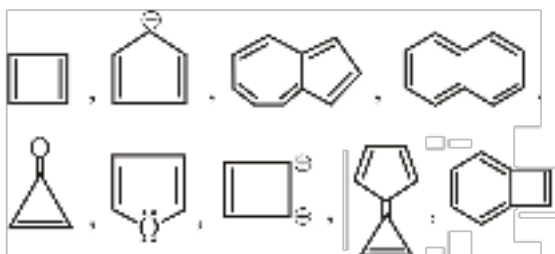


- (1) All three C-N bond length are same.
- (2) C₁-N and C₃-N bonds length are same but shorter than C₅-N bond length.
- (3) C₁-N and C₅-N bonds length are same but longer than C₃-N bond length.
- (4) C₁-N and C₃-N bonds length are different but both are longer than C₅-N bond length.

35) Among the given sets, which represents the resonating structures ?

- (1) $\text{H}-\text{C}\equiv\text{N}-\ddot{\text{O}}^-$ and $\text{H}-\ddot{\text{O}}-\text{C}\equiv\text{N}:$
- (2) $\text{H}-\ddot{\text{O}}^+=\text{C}=\ddot{\text{N}}^-$ and $\text{H}-\ddot{\text{O}}-\text{C}\equiv\text{N}:$
- (3) $\text{H}-\text{C}\equiv\text{N}^+-\ddot{\text{O}}^-$ and $\text{H}-\text{C}(=\text{O})-\ddot{\text{N}}:$
- (4) $\text{H}-\ddot{\text{O}}-\text{C}=\ddot{\text{N}}:$ and $\text{H}-\ddot{\text{N}}=\text{C}=\ddot{\text{O}}:$

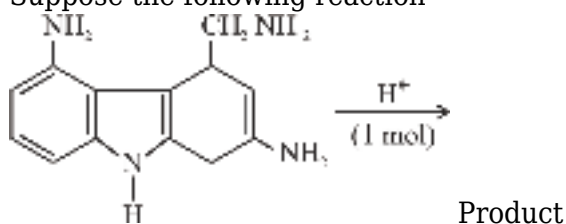
36) Number of aromatic compounds in given compounds :-



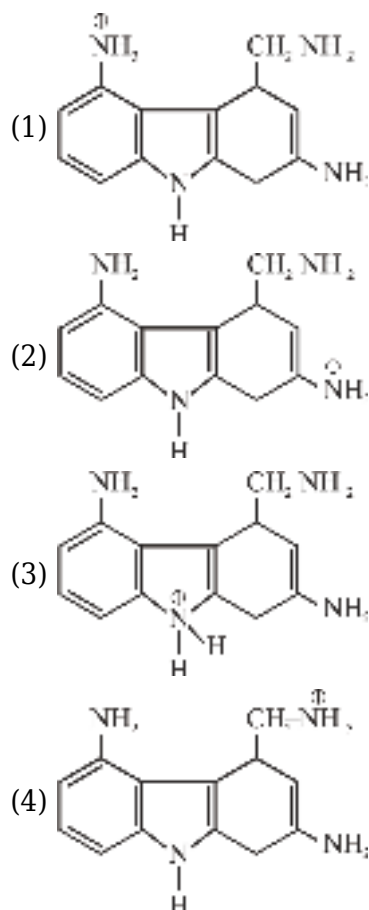
- (1) 6
- (2) 7
- (3) 4
- (4) 5

37)

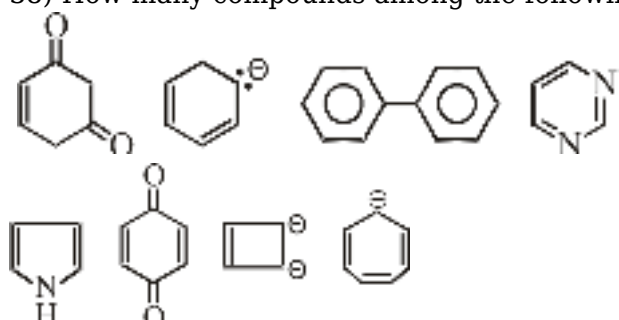
Suppose the following reaction -



The main product of reaction will be :-



38) How many compounds among the following are aromatic?



- (1) 3
(2) 5
(3) 6
(4) 4

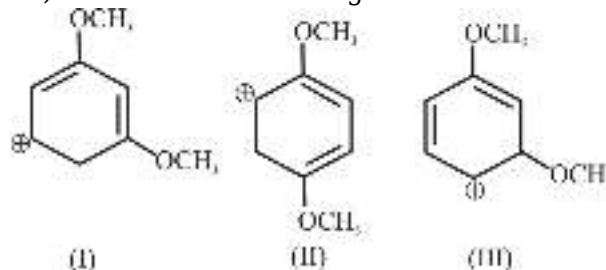
39) Match list-I with list-II

	List-I (Compound)		List-II (pK _b -value)
(a)	Benzenamine	(i)	4.70
(b)	Phenylmethanamine	(ii)	8.92
(c)	N-Methylaniline	(iii)	9.38

(d)	N,N-Dimethylaniline	(iv)	9.30
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- (1) a-(iii), b-(i), c-(iv), d-(ii)
 (2) a-(iii), b-(iv), c-(i), d-(ii)
 (3) a-(iv), b-(iii), c-(ii), d-(i)
 (4) a-(ii), b-(iii), c-(i), d-(iv)

40) Which of the following order is correct for the stability of given carbocation :-



- (1) I > II > III
 (2) III > II > I
 (3) I > III > II
 (4) II > III > I

41) How many groups show +M effect

$-\text{NO}_2$, $-\text{COOCH}_3$, $-\text{NH}-\text{COCH}_3$, $-\text{F}$,

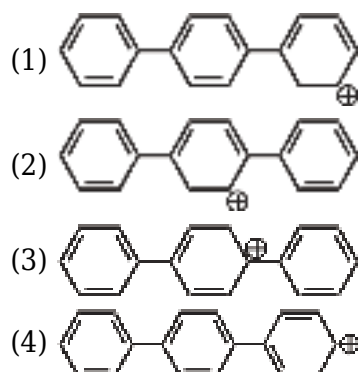
$-\text{O}-\text{CH}_3$, $-\text{CH}_2^+$, $-\text{NR}_3^+$, $-\text{CONH}_2$, $-\text{OCOCH}_3$,

$-\text{CH}_2^-$, $-\text{CH}_3$, $-\text{C}(\text{CH}_3)_2$, $-\text{C}(=\text{O})-\text{O}^-$,

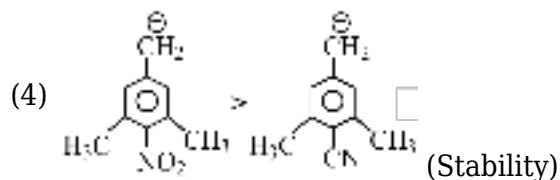
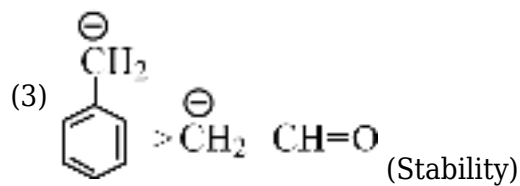
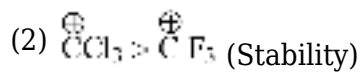
$-\text{CONH}-\text{CH}_3$

- (1) 5
 (2) 6
 (3) 4
 (4) 7

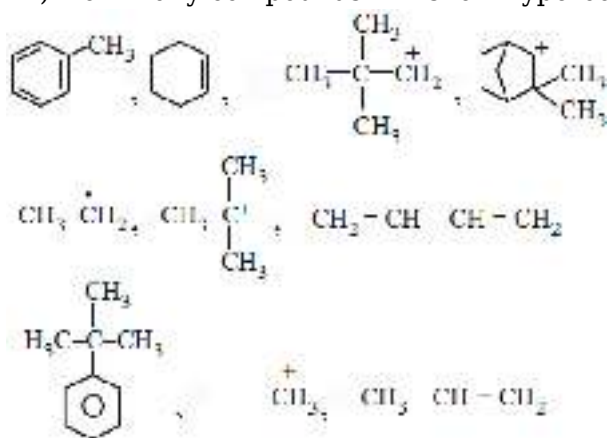
42) Most stable carbocation :-



43) Correct order is ?



44) How many compounds will show hyperconjugation effect



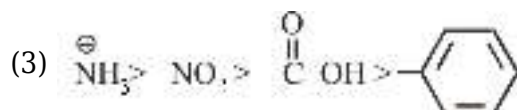
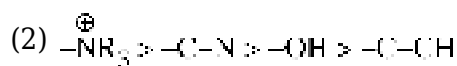
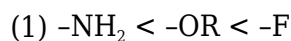
(1) 5

(2) 6

(3) 7

(4) 8

45) The correct order of -I effect is :-



(4) All of the above

BIOLOGY

1) Young one of the cockroach is called

(1) Naid

(2) Grub

- (3) Nymph
- (4) Maggot

2) In cockroach a sac like structure present in alimentary canal and used for storing of food called :-

- (1) Pharynx
- (2) Gizzard
- (3) Crop
- (4) Stomach

3) Which structure represent the brain of cockroach?

- (1) Supra-Pharyngeal ganglion
- (2) Sub-Pharyngeal ganglion
- (3) Supra-Oesophageal ganglion
- (4) Sub-Oesophageal ganglion

4) The head of the cockroach is formed by the fusion of how many segments-

- (1) Five
- (2) Six
- (3) Three
- (4) Ten

5) Incorrect about cockroach-

- (1) The adults of the common species of cockroach, *Periplaneta americana* are about 34-53 cm long with wings that extend beyond the tip of the abdomen in males.
- (2) The body of the cockroach is segmented and divisible into three distinct regions - head, thorax and abdomen.
- (3) The entire body is covered by a hard chitinous exoskeleton (brown in colour).
- (4) Sclerites are joined to each other by a thin and flexible articular membrane (arthrodial membrane).

6) Main difference between male and female cockroach is of-

- (1) Antennae
- (2) Mandible
- (3) Podomeres
- (4) Anal style

7) In cockroach, the ootheca is formed by the secretion of :-

- (1) Phallic gland
- (2) Uricose gland
- (3) Collateral gland

(4) Mushroom gland

8) Choose the statement which is not correct for cockroach?

- (1) Blood vascular system of cockroach is a closed type.
- (2) The abdomen in both males and females consists of 10 segments.
- (3) On an average female produce 9-10 oothecae.
- (4) Malpighian tubules help in excretion.

9) Spiracles found in cockroach are :-

- (1) 2 pairs in thorax and 8 pairs in abdomen
- (2) 2 pairs in thorax and 10 pairs in abdomen
- (3) 8 pairs in thorax and 2 pairs in abdomen
- (4) 2 pairs in thorax and 4 pairs in abdomen

10) **Assertion** : Frogs exhibit sexual dimorphism.

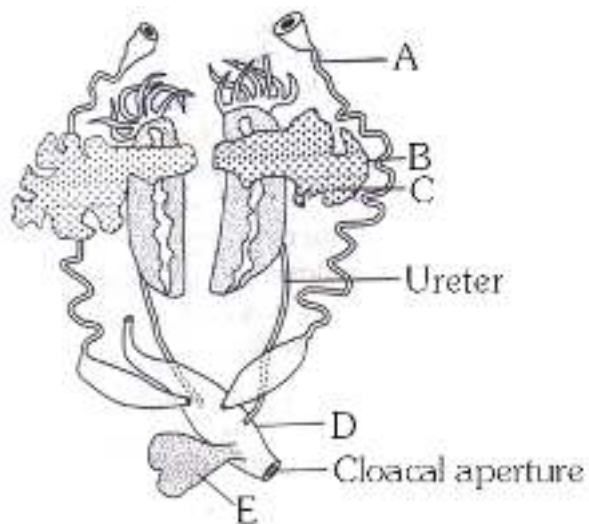
Reason : Male frog can be distinguished by the presence of sound producing vocal sacs which is absent in female frogs.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

11) Given below is a list of some structure : Diaphragm, Tail, Anus, Cloaca, Nictitating membrane and the cortex and medulla in kidney. How many of the above structure are not found in *Rana tigrina* ?

- (1) 3
- (2) 5
- (3) 6
- (4) 4

12) Observe the following figure of female reproductive system of frog and identify A to E.



- (1) A-Urinary duct, B-Ova, C-Ovary,
D-Cloaca, E-Urethra
- (2) A-Oviduct, B-Ovary, C-Ova, D-Cloaca,
E-Urinary bladder
- (3) A-Oviduct, B-Ovary, C-Ova, D-Rectum,
E-Adrenal gland
- (4) A-Urinogenital duct, B-Ovary, C-Ovum,
D-Coelom, E-Urethra

13) Which Statement is true (about frogs) :

- (1) Kidney is compact, light red and bean shape structures situated a little posteriorly in the body cavity on both sides of vertebral column
- (2) Food is captured by the trilobed tongue. Digestion of food takes place by the action of HCl and gastric juices secreted from the walls of the stomach
- (3) Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the second digit of the fore limbs which are absent in female frogs
- (4) They have the ability of change the colour to hide them from their enemies. This protective coloration is called mimicry.

14) Frog has different types of sense organs.

- (I) Sensory papillae
- (II) Nasal epithelium
- (III) Taste buds
- (IV) Eyes
- (V) Tympanum with internal ears

Which of these are well organized structures?

- (1) I and III
- (2) III and IV
- (3) IV and V
- (4) I, II, III and IV

15) Which of the following respiration take place during hibernation of frog :

- (1) Pulmonary respiration
- (2) Cutaneous respiration
- (3) Buccal cavity respiration
- (4) Gill respiration

16) In frog, a triangular structure calledA..... Joins the right atrium the ventricle opens into a sac likeB.... on the ventral side of heart. Here A and B are respectively.

- (1) A → Sinus Venosus; B → Conus arteriosus
- (2) A → Conus arteriosus; B → Sinus Venosus
- (3) A → Truncus arteriosus; B → Sinus Venosus
- (4) A → Sinus Venosus; B → Systemic aorta

17) An epithelium tissue which has thin flat cells, appears like closely packed tiles is found to be present at ?

- (1) Inner lining of stomach
- (2) Inner lining of fallopian tube
- (3) Inner lining of urinary bladder
- (4) Bowman's Capsule

18) Stratified squamous Epithelium :-

- (1) Outer most layer squamous & Inner most is cuboidal
- (2) Outer most layer cuboidal & Inner most is squamous
- (3) Outer most layer columnar & Inner most is cuboidal
- (4) Outer most layer cuboidal & Inner most is columnar

19) Which of the following are **correct** for epithelial tissue?

- (A) It has free surface which may face towards the body fluid.
- (B) It provides covering for some part of body
- (C) Intercellular matrix is abundant
- (D) It lies on basement membrane.

- (1) (A), (B), (C), (D)
- (2) Only (C)
- (3) (A), (C), (D)
- (4) (A), (B), (D)

20) Tendons are :-

- (A) Dense regular connective tissue with elastic fibres.
- (B) Dense regular connective tissue with collagen fibres.
- (C) Dense irregular connective tissue with collagen fibres.
- (D) Dense irregular connective tissue with elastic fibres.

- (1) Only (B)
- (2) (B) and (C)

(3) (A) and (D)

(4) (A) and (C)

21) Which of the following statement is correct.

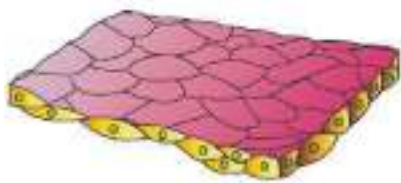
(1) matrix of cartilage semi solid, pliable and can resist compression

(2) Bone is mineralised, solid connective tissue having matrix rich in calcium salt and collagen fibers

(3) matrix of cartilage is secreted by osteocytes

(4) Tendon connects bone to bone

22) Identify the type of epithelial tissue shown below as well as the related right place of its occurrence in our body along with its correct function and select the correct option for the two together ?



	Type of epithelial Tissue	Occurance & its function
(1)	Simple squamous Epithelium	- Found in wall of blood vessels and air sacs of lungs. - Function-forming a diffusion boundary
(2)	Simple cuboidal epithelium	- Found in wall of ducts of gland - Function-diffusion and excretion
(3)	Simple columnar epithelium	- Found in wall of nephrones and wall of ducts of glands - Function-secretion and Absorption
(4)	Simple squamous epithelium	- Found in wall of stomach and intestine - Function-absorption and secretion

(1) 1

(2) 2

(3) 3

(4) 4

23) It serves as a support framework for epithelium. It contains fibroblast, macrophages and mast cells. It is known as.

- (1) Dense regular tissue
- (2) Adipose tissue
- (3) Dense irregular tissue
- (4) Areolar tissue

24)

Respiratory organ arthropoda is :-

- (1) Gills
- (2) Book gills
- (3) Book lungs
- (4) 1,2 & 3

25) Which annelid is dioecious.

- (1) Earthworm
- (2) Leech
- (3) Nereis
- (4) Ascaris

26) An animal phylum having radially symmetrical adults but bilateral symmetrical larvae is :

- (1) Porifera
- (2) Coelenterata
- (3) Echinodermata
- (4) Annelida



27) Which of the following features is not correct for the above creature :

- (1) Comb plates (External rows)
- (2) Cnidoblast
- (3) Sexes are not separate
- (4) Digestion is both extracellular and intracellular

28) The body is externally and internally divided into segments with a serial repetition of at least some organs in

- (1) Hydra
- (2) Fasciola

(3) Earthworm

(4) Pila

29) Which of the following is limbless amphibian ?

(1) Hyla

(2) Bufo

(3) Salamandra

(4) Ichthyophis

30) Vertebrata is a subphylum and Pisces is a

(1) Class

(2) Super class

(3) Division

(4) Sub class

31) A zoologist came across a fish that has to swim constantly in water to avoid sinking. Identity that particular fish.

(1) Fighting fish

(2) Dog fish

(3) Angel fish

(4) Catla

32) Polyp phase is absent in.....

(1) Hydra

(2) Aurelia

(3) Physalia

(4) Obelia

33) When any longitudinal plane passing through the central axis of the body divides the organism into two identical halves, it is called-

(1) Radial symmetry

(2) Bilateral symmetry

(3) Biradial symmetry

(4) Asymmetry

34) The long bones are hollow and connected by air passage. They are the characteristics of

(1) Aves

(2) Mammalia

(3) Reptilia

(4) Land vertebrates

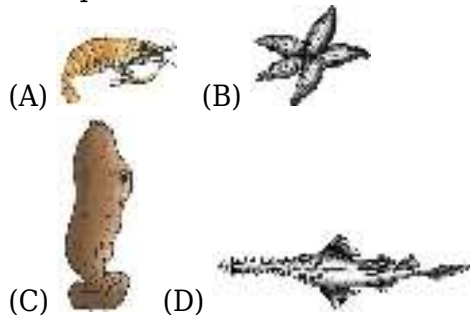
35) Some animals are given in the list below :

Asterias, Pila, Echinus, Antedon, Pinctada, Cucumaria, Octopus, Loligo, Ophiura, Dentalium

How many animals among these are related to the second largest phylum of animals?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

36) Examine the given table with animal taxon is correctly match their characters with suitable examples :



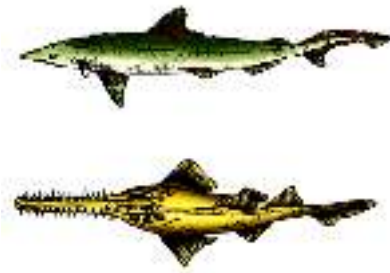
	A	B	C	D
(1)	Arthropoda rasping organ radula is present Ex. Prawn	Echinodermata spiny skin Ex. <i>Asterias</i>	Urochordata Marine animal Ex. <i>Ascidia</i>	Osteichthyes Bony endoskeleton Ex. <i>Scoliodon</i>
(2)	Arthropoda Chitinous exoskeleton Ex. Prawn	Echinodermata Sexual reproduction Ex. <i>Asterias</i>	Urochordata notochord is present in larval tail Ex. <i>Ascidia</i>	Chondrichthyes Cartilagenous endoskeleton Ex. - <i>Pristis</i>
(3)	Annelida Segmented body Ex. Scorpion	Mollusca Calcereous shell Ex. <i>Ophiura</i>	Cephalochordata Marine animal Ex. <i>Salpa</i>	Amphibia Oviparous Ex. <i>Salamendra</i>
(4)	Aschelminthes Common name roundworm Ex. Prawn	Arthropoda Malpighian tubules present Ex. Butter fly	Hemichordata Proboscis gland present Ex. <i>Saccoglossus</i>	Osteichthyes Operculum present Ex. <i>Labeo</i>

- (1) 1
- (2) 2
- (3) 3
- (4) 4

37) **Assertion :-** Hippocampus is an example of subphylum vertebrata.

Reason :- The members of subphylum vertebrata possess notochord during the embryonic period.

- (1) Assertion is correct but reason is incorrect
- (2) Both assertion and reason are incorrect
- (3) Both assertion and reason are correct but reason is not the correct explanation of assertion
- (4) Both assertion and reason are correct but reason is the correct explanation of assertion



38) Identify the correct statement w.r.t given figure:

- (1) In females pelvic fins bear claspers
- (2) Skin is covered by cycloid scales
- (3) They are mostly oviparous
- (4) Mouth is located ventrally

39) Select the incorrect statement regarding arthropoda ?

- (1) They have organ system level of organisation
- (2) Circulatory system is open type
- (3) They are pseudocoelomate animals
- (4) They have jointed appendages

40) **Statement-I** : Body of *Saccoglossus* divided into anterior proboscis, collar and long trunk.

Statement-II : *Balanoglossus* show indirect development and organ system level of organization.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect
- (3) Only statement-I is correct.
- (4) Only statement-II is correct.

41) Match the column-I with the column-II and find out the correct answer :

	Column-I		Column-II
(A)	Proboscis gland	(i)	<i>Aurelia</i>
(B)	Cnidocytes	(ii)	<i>Psittacula</i>
(C)	Electric organs	(iii)	<i>Balanoglossus</i>
(D)	Air sacs	(iv)	<i>Torpedo</i>

- (1) A-iii, B-i, C-ii, D-iv
- (2) A-i, B-iii, C-iv, D-ii
- (3) A-iii, B-i, C-iv, D-ii
- (4) A-ii, B-i, C-iii, D-iv

42) There are some features of animals are given :-

Organ system level, Coelomate, Open circulation, Triploblastic, Unsegmented, Intracellular digestion.

How many characters are present in the members of phylum arthropoda -

- (1) 3
- (2) 4
- (3) 5
- (4) 2

43) What is the right sequence of structure of mollusca from outside to inside -

- (1) Shell → Mantle cavity → Mantle → Visceral hump.
- (2) Mantle → Shell → Mantle cavity → Visceral hump.
- (3) Shell → Mantle → Mantle cavity → Visceral hump.
- (4) Shell → Mantle → Visceral hump. → Mantle cavity

44) Which one of the following statement is totally wrong about the occurrence of notochord while the other three are correct ?

- (1) It is present throughout life in Amphioxus
- (2) It is present only in larval tail in Ascidians
- (3) It is replaced by a vertebral column in adult frog
- (4) It is absent throughout life in humans from the very beginning of embryological stage

45) **Statement-I** : Ctenophora show bioluminescence property and only reproduce sexually.
Statement-II : *Planaria* possess high regeneration capacity and show internal fertilization.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect
- (3) Only statement-I is correct.
- (4) Only statement-II is correct.

46) Which of the following is correctly matched?

- (1) *Funaria*: Water is required for fertilization
- (2) *Adiantum*: Produce two different types of spores
- (3) *Chlamydomonas*: Unicellular alga classified in kingdom Plantae by Whittaker
- (4) *Volvox*: Colonial blue green algae

47) Chlorophyll a and chlorophyll b are commonly present in:

- (1) Green algae
- (2) Blue green algae
- (3) Red algae
- (4) 1 and 2 both

48) Match the Column I with Column II

Column I	Column II
----------	-----------

A.	<i>Chlorella</i>	P.	Naked seed plant
B.	<i>Equisetum</i>	Q.	Peat Moss
C.	<i>Sphagnum</i>	R.	Homosporous Pteridophyta
D.	<i>Ginkgo</i>	S.	Unicellular green algae

(1) A: P; B: Q; C: R; D: S

(2) A: S; B: R; C: Q; D: P

(3) A: S; B: R; C: P; D: Q

(4) A: R; B: Q; C: P; D: S

49) Male and female reproductive structures are present in different plants in:

(1) *Pinus and Cycas*

(2) *Funaria and Marchantia*

(3) *Cycas and Marchantia*

(4) *Chara and Marchantia*

50) Which of the following is not associated with all seeded plant?

(1) fruit formation

(2) Heterospory

(3) Seed formation

(4) Independent sporophyte

51) Antheridium is present in:

(1) Algae and gymnosperm

(2) Algae and Bryophyta only

(3) Algae, Bryophyta and Pteridophyta

(4) Bryophyta, Pteridophyta and Gymnosperm

52) Large size leaves with homospory are observed in:

(1) *Pteris*

(2) *Equisetum*

(3) *Selaginella*

(4) *Salvinia*

53) **Assertion (A):** *Marchantia* is a dioecious plant while *Funaria* is a monoecious plant

Reason (R): Male and female reproductive organs archegonium and antheridium respectively are present on different plant in *Marchantia* while they are on same plant in *Funaria*

(1) A and R both are true and R explain A

(2) A is true but R is false

(3) A and R both are true and R does not explain A

(4) A and R both are false

54) Pinnately compound leaves are commonly observed in:

- (1) *Adiantum*
- (2) *Cycas*
- (3) *Adiantum* and *Cycas* both
- (4) *Pinus*

55) Which of the following set of plant produce non motile gametes ?

- (1) *Spirogyra*, *Pinus* and *Cycas*
- (2) *Spirogyra*, *Porphyra* and *Sphagnum*
- (3) *Pinus*, *Pisum* and *Spirogyra*
- (4) *Porphyra*, *Nostoc* and *Pinus*

56) Endosperm of gymnosperm is:

- (1) Male gametophyte
- (2) Female Sporophyte
- (3) Female gametophyte
- (4) Megasporangia

57) Which of the following is not associated with red algae ?

- (1) Pre dominance of r-phycoerythrin
- (2) Fusion of gametes inside oogonium
- (3) Present only in greater depth of water
- (4) Some members are used as food

58) Evolutionary first plants to have embryo are:

- (1) Bryophyta
- (2) Pteridophyta
- (3) Gymnosperm
- (4) Algae

59) Antheridium and archegonium in Bryophyta and Pteridophyta are:

- (1) Multicellular and unicellular respectively
- (2) Multicellular and Multicellular respectively
- (3) unicellular and unicellular respectively
- (4) Multicellular and unicellular respectively

60) Few stages of life cycle of Pteridophyta are given below. Arrange these events in correct order starting from sex organ bearing structure and select the correct option from option given below

A: Embryo
B: Prothallus
C: Spore formation
D: Spore germination
E: Fertilization
F: Main plant

- (1) $B \rightarrow A \rightarrow C \rightarrow D \rightarrow E \rightarrow F$
- (2) $B \rightarrow E \rightarrow A \rightarrow F \rightarrow D \rightarrow C$
- (3) $F \rightarrow C \rightarrow D \rightarrow B \rightarrow E \rightarrow A$
- (4) $B \rightarrow E \rightarrow A \rightarrow F \rightarrow C \rightarrow D$

61) Two statements 1 and 2 are given below:

Statement 1 : Leaves of gymnosperms are well adapted for extremes of environment

Statement 2 : Needle like leaves of conifers reduces the surface area and reduces the transpiration loss of water

- (1) Statement 1 and 2 both are false
- (2) Only statement 1 is true
- (3) Only Statement 2 is true
- (4) Statement 1 and 2 both are true

62) Zygotes do not undergo reduction division immediately and produce a multicellular body called a sporophyte in:

- (1) Bryophyta only
- (2) Bryophyta and Pteridophyta only
- (3) Algae, Bryophyta and Pteridophyta only
- (4) Bryophyta, Pteridophyta and Gymnosperm

63) Which of the following brown algae show isogamy?

- (1) *Ectocarpus*
- (2) *Fucus*
- (3) *Sargassum*
- (4) 1 and 2 both

64) Which of the following is not associated with *Sphagnum*?

- (1) It is used as packing material for trans-shipment of non-living material
- (2) It has independent gametophyte
- (3) Multicellular rhizoids are present
- (4) It is a monoecious plant

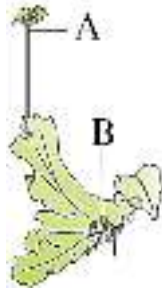
65) Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous. The megaspores and microspores germinate and give rise to female and male gametophytes, respectively.

Read the following statements A and B carefully and select the correct option :

Statement A : The female gametophytes in these above plants are retained on the parent sporophytes for variable periods.

Statement B : The development of the zygotes into young embryos take place within the male gametophytes.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Statement A and B both are correct
- (4) Statement A and B both are incorrect



66) identify A and B in given figure :

- (1) A: Archegonia, B: Gemma cup
- (2) A: Antheridium, B: Gemma cup
- (3) A: Archegoniophore, B: Gemma cup
- (4) A: Antheridiophore, B: Gemma cup

67) Male and female gametes in *Cycas* are

- (1) Motile and non-motile respectively
- (2) Non motile and motile respectively
- (3) Both are motile
- (4) Both are non-motile

68) Gametophytic and sporophytic both generations are independent to each other in :

- (1) *Salvinia*
- (2) *Pteridium*
- (3) *Funaria*
- (4) *Pinus*

69) A collection of species, which bears a close resemblance to one another in the morphological characters of the floral parts is known as :-

- (1) Family
- (2) Genus
- (3) Division
- (4) Variety

70) *Panthera leo*, *Panthera pardus* and *Panthera tigris* represent :-

- (1) They are member of same species
- (2) They are species of different genus
- (3) They are different species of same genus
- (4) *Panthera* is a name of species while *leo*, *pardus* and *tigris* represent variety

71)

Both the words in a biological name when handwritten are underlined or printed in italics to indicate :-

- (1) They are endangered
- (2) They are living
- (3) Their Latin origin
- (4) Now they are extinct

72) How many of the following terms are not related with classification of Mango ? *Mangifera*, Anacardiaceae, Poales, Dicotyledonae, Angiospermae, Monocotyledonae, Polymoniales, Convolvulaceae, Plantae

- (1) One
- (2) Four
- (3) Three
- (4) Two

73)

Consider the following four statements (a-d) and select the option which include all the correct ones only regarding Ernst Mayr :

- (a) The Darwin of the 20th century.
 - (b) He is pioneered the currently accepted definition of a biological concept of species
 - (c) Mayr was awarded nobel prize in the field of Evolutionary biology
 - (d) His research spanned Ornithology, Taxonomy, systematics, Zoogeography and Evolution.
- (1) Statements : (a) and (b) only
 - (2) Statements : (a), (b) and (d)
 - (3) Statements : (b), (c) and (d)
 - (4) Statements : (a), (b), (c) and (d)

74) Study the given statements carefully and give the answer :-

- (A) A genus is a group of related species which has more characters in common in comparison to species of other genera
- (B) Families are characterised on the basis of both vegetative and reproductive features of plants species.
- (C) Felidae and canidae are the families of cat and dog respectively
- (D) To determine relations in higher taxa is more difficult in comparison to lower taxa.

How many statements are correct ?

- (1) One

- (2) Four
- (3) Three
- (4) Two

75) What is incorrect for chemosynthetic autotrophic bacteria :-

- (1) Oxidise various organic and inorganic substances
- (2) Use the energy released through oxidation for their ATP production
- (3) Play great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur
- (4) They are most abundant in nature

76) Study the following statements :

- (I) All prokaryotic organisms were grouped together under Kingdom fungi.
- (II) The unicellular eukaryotic organisms were placed in Kingdom protista.
- (III) *Chlorella* and *Chlamydomonas* both have cell wall.
- (IV) *Paramoecium* and *Amoeba* lack cell walls.
- (V) Kingdom protista has brought together *Chlamydomonas* , *Chlorella* with *Paramoecium* and *Amoeba*.

Which of the statements given above are correct according to Whittaker classification ?

- (1) II, III and IV only
- (2) II, III, IV and V only
- (3) I, II, III and IV only
- (4) I, II, III, IV and V

77) How many of the following fungi have septate and Branched Mycelium with absence of Asexual spores

Neurospora, Alternaria, Claviceps, Agaricus, Colletotrichum, Penicillium, Trichoderma, Aspergillus, Mucor, Rhizopus and Albugo

- (1) 8
- (2) 3
- (3) 1
- (4) All

78) A proteinaceous covering which is elastic enough to enable turning and flexing of the cell, yet rigid enough to prevent excessive alternation in shape is :-

- (1) Pellicle in *Euglena*
- (2) Silica in diatoms
- (3) Budding in yeast
- (4) Plasma membrane in *Gonyaulax*

79) Identify the correct match from column-I and II :

	Column I		Column II
--	----------	--	-----------

(i)	Euglena	(a)	Saprophytic
(ii)	Diatom	(b)	Diatomaceous earth
(iii)	Slime mould	(c)	Red tide
(iv)	Dinoflagellates	(d)	<i>Gonyaulax</i>
		(e)	Pellicle
		(f)	Chief producer of oceans

- (1) (i)-e, (ii)-b,f (iii)-c, (iv)-a,d
 (2) (i)-e, (ii)-b,f (iii)-a, (iv)-c,d
 (3) (i)-e, (ii)-b,c,f (iii)-a, (iv)-d
 (4) (i)-e, (ii)-b,c,f (iii)-d, (iv)-a

80) Select the wrong statement :

- (1) Diatoms are photosynthetic
 (2) Diatoms are microscopic and float passively in water
 (3) The walls of diatoms are easily destructible
 (4) 'Diatomaceous earth' is formed by the cell walls of diatoms.

81) **Assertion** : Mesosome increases the surface area of plasma membrane for enzymatic content.

Reason : Mesosomes are foldings of plasma membrane towards cell wall.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 (3) Assertion is true but Reason is false.
 (4) Both Assertion and Reason are false.

82) Which one of the following is exclusively found in blue green algae but not found in any other prokaryotes :-

- (1) Peptidoglycan's cell wall
 (2) Chlorophyll 'a'
 (3) Lipoproteins cell membrane
 (4) Nitrogen fixation ability

83) **Statement -I** : Mycoplasma is smallest living organism that can survive without oxygen.

Statement -II : Members of protista are primary aquatic.

- (1) Both the statements are correct.
 (2) Both the statements are incorrect.
 (3) Statement-I is correct whereas statement-II is incorrect.
 (4) Statement-I is incorrect whereas statement-II is correct.

84) **Assertion** :- Archaeobacteria live in extremely adverse conditions such as very high temperature and high salt concentration.

Reason :- Archaeobacteria differ from other bacteria in having a different cell wall structure and complex cell membrane.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

85) According to five kingdom system scheme, which statement is common about given organism :-
Euglena, Nostoc, Chlorella, Diatoms and Dinoflagellates

- (1) All are unicellular eukaryotes
- (2) All are multicellular eukaryotic organism
- (3) All are photosynthetic unicellular prokaryotes
- (4) They are photosynthetic and act as producer in nature

86)

Select the option which includes all the correct ones only :-

- (A) Diatoms → Chief producer of ocean
- (B) Dinoflagellate → Without flagella
- (C) Slime mould → Decomposer nature
- (D) Euglenoids → Sometime behave like predator

(E) Protozoa → Unicellular prokaryotes

- (1) A, B and C
- (2) A, B, C and D
- (3) B, C, D and E
- (4) A, C and D

87) Read the following statement (A-D) (A) Cell wall of Fungi consists of chitin.

(B) Fungi are heterotrophic

(C) Fungi can also live as symbionts in association with algae as lichens and with roots of higher plants as mycorrhiza

(D) Fusion of two nuclei called plasmogamy

How many of the above statement are correct ?

- (1) Two
- (2) Three
- (3) Four
- (4) One

88) Read the given below statements and choose correct answer -

Statement-I : *Albugo* do not show karyogamy immediately after plasmogamy. In *Albugo* an intervening dikaryotic stage ($n + n$, i.e. two nuclei per cell) occurs : such a condition is called a dikaryon and the phase is called dikaryophase.

Statement-II : *Neurospora* and *Puccinia*, show karyogamy immediately after plasmogamy.

- (1) Both statement I and statement II are correct.
- (2) Both statement I and statement II are incorrect.
- (3) Statement I is correct and statement II is incorrect.
- (4) Statement I is incorrect and statement II is correct.

89) Choose the incorrect pair of following :-

- (1) Phycomycetes-aquatic habitat
- (2) Ascomycetes-Rust and smut disease
- (3) Basidiomycetes-lack sex organs
- (4) Deuteromycetes-help in mineral cycling

90) Select the correct pair :-

- (1) Lichens - Grow in Polluted areas.
- (2) Prions - Abnormally folded protein
- (3) Virus - Cause Potato spindle tuber disease
- (4) Viroids - Consist of protein and nucleic acid

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	1	1	2	1	2	2	1	3	1	2	4	1	4	2	1	3	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	4	1	3	4	4	4	4	4	3	4	1	4	1	4	4	3	2	3
Q.	41	42	43	44	45															
A.	2	3	1	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	3	2	1	4	1	3	1	1	4	4	3	2	2	4	4	4	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	2	3	1	1	2	2	4	3	3	2	3	2	1	4	2	1	1
Q.	86	87	88	89	90															
A.	1	1	1	1	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	2	1	4	3	1	1	1	1	2	4	3	2	1	4	1	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	4	4	3	3	3	3	4	2	2	2	1	1	4	2	3	4	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	3	4	1	1	1	2	3	1	3	1	2	3	3	3	3	1	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	1	1	1	4	1	2	2	3	3	2	2	2	4	2	3	1	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	1	1	4	4	2	2	2	2										

SOLUTIONS

PHYSICS

1)

Since ammeter is ideal, its resistance is zero
its mean reading of volt meter is also zero

$$2) I = \frac{4-1}{3} = 1A$$

Pot. difference of A = $E - Ir = 4 - 1 \times 1 = 3V$

Pot. difference of B = $E + Ir = 1 + 1 \times 1 = 2V$

$$P = I^2 R = 1 \times 1 = 1W$$

$$V = IR = 1 \times 1 = 1 \text{ volt}$$

$$3) I = \frac{4+12}{8} = 2A$$

4) Balancing length does not depend on position of battery and galvanometer

$$5) S = \frac{I_g R_g}{I - I_g} = \frac{10 \times 10^{-3} \times 100}{10} = 0.1 \Omega$$

which is connected in parallel to galvanometer

$$6) v = i_g (G+S)$$

7) For maximum energy equivalent resistance of combination should be minimum.

8)

By KVL

$$V_A - 2 \times 2 - 3 - 2 \times 1 = V_B$$

$$V_A - V_B = 9V$$

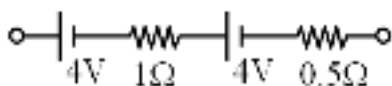
$$9) \frac{5S}{R} = \frac{20}{80}$$

$$R = 220 \Omega$$

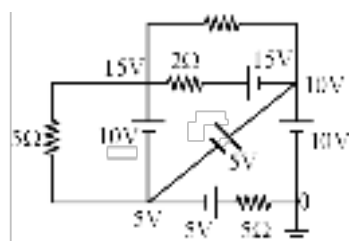
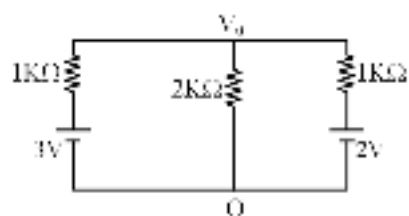
$$R_{AB} = 10\Omega$$

Equivalent emf of 2 V and 6 V

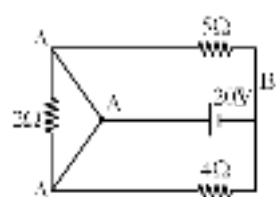
$$E^1 = \frac{2 \times 1 + 6 \times 1}{1 + 1} = 4V$$



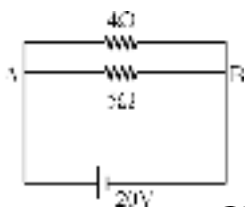
$$E_{\text{net}} = 8 \text{ V}$$


$$V_x = 15 \text{ V}$$

$$V^0 = \frac{\frac{3}{1} + 0 + \frac{2}{1}}{\frac{1}{1} + \frac{1}{1} + \frac{1}{2}} = 2V$$

Resistivity remains constant



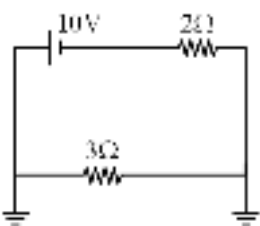
current in $2\Omega = 0$



$$\text{current in } 4\Omega = \frac{20}{4} = 5 \text{ A}$$

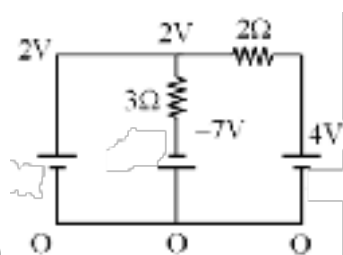
$$\text{current in } 5\Omega = \frac{20}{5} = 4 \text{ A}$$

correct ans. a, b, c, d



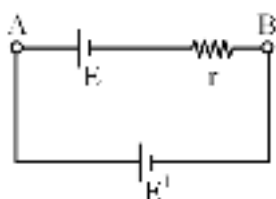
16) Since current in 3Ω is zero

$$I = \frac{10}{2} = 5 \text{ A}$$



17)

$$I = \frac{9}{3} = 3 \text{ A}$$



18)

If $E^1 > E$ then from A to B

If $E^1 < E$ then from B to A

If $E^1 = E$ then $I = 0$

19) $\vec{P} \cdot \vec{Q} = 0$

$$a^2 - 2a - 3 = 0$$

$$a^2 - 3a + a - 3 = 0$$

$$a(a - 3) + 1(a - 3) = 0$$

$$(a - 3)(a + 1) = 0$$

$$\boxed{a = 3}$$

$$20) \cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1$$

$$\cos^2 \alpha = \frac{1}{3}$$

$$\cos \alpha = \frac{1}{\sqrt{3}}$$

$$\text{so for x axis } \frac{1}{\sqrt{3}} = \frac{r_x}{30} \Rightarrow 10\sqrt{3} \text{ units}$$

same for other axes also

$$21) R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$17 = \sqrt{8^2 + 15^2 + 2(15)(8) \cos \theta}$$

$$289 = 289 + 240 \cos \theta$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

$$22) \vec{A} + \vec{B} = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

$$= \frac{(\vec{A} + \vec{B}) \cdot \vec{B}}{|\vec{B}|} = \frac{3 + 6 + 6}{\sqrt{1 + 4 + 4}} = \frac{15}{3} = 5$$

$$23) \vec{A} \times \vec{B} = \sqrt{3} \vec{A} \cdot \vec{B}$$

$$AB \sin \theta = \sqrt{3} AB \cos \theta$$

$$\tan \theta = \sqrt{3}$$

$$\theta = 60^\circ$$

$$24) \vec{P} = 7\hat{i} + 6\hat{j}$$

$$\vec{P} + \vec{Q} = 11\hat{i} + 9\hat{j}$$

$$7\hat{i} + 6\hat{j} + \vec{Q} = 11\hat{i} + 9\hat{j}$$

$$\vec{Q} = 4\hat{i} + 3\hat{j}$$

$$|\vec{Q}| = 5$$

$$25) \frac{dy}{dx} = 8x + 2 \text{ Now if we take as 'y' then it is a form of } y = mx + c \text{ i.e. straight line}$$

$$V_{\text{avg}} = \frac{\int_0^1 V dt}{\int_0^1 dt} = \frac{\int_0^1 (2t + 3) dt}{\int_0^1 dt} = \frac{(t^2 + 3t)_0^1}{1}$$

$$26) \\ = 4 \text{ m/s}$$

$$27) U + \frac{1}{2}a(2n-1) = 3 \left[\frac{1}{2}g(3)^2 \right]$$

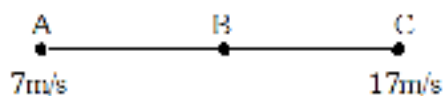
$$U - \frac{1}{2}g(2-1) = 135$$

$$U - 5 = 135$$

$$U = 140 \text{ m/s}$$

$$28) v^2 - u^2 = 2as$$

$$(17)^2 - (7)^2 = 2 \times a s$$



$$v^2 - (7)^2 = as$$

$$v^2 - (7)^2 = \frac{(17)^2 - (7)^2}{2}$$

$$(v^2 - (7)^2) = 24 \times 10$$

$$v^2 = 120 + 49 = 169$$

$$v_B = 13$$

$$13 = 7 + at_1 \Rightarrow at_1 = 6$$

$$17 = 13 + at_2 \Rightarrow at_2 = 4$$

$$t_1 : t_2 = 3 : 2$$

Uniform acceleration

$$v_{avg} = \frac{v_1 + v_2}{2}$$

$$29) \frac{dv}{dt} = a = \text{constant}$$

$$30) x^2 = \frac{4}{3}t$$

Differentiation with respect to t

$$2x \frac{dx}{dt} = \frac{4}{3}$$

$$2xv = \frac{4}{3}$$

$$v = \frac{2}{3x}$$

$$\text{At } x = \frac{1}{6} \text{ m} \Rightarrow v = \frac{2}{3(1/6)} = 4 \text{ m/s}$$

$$31) x = at^2 + bt \dots (1) \quad (a > 0, b < 0)$$

$$v = 2at + b \dots (2)$$

$$a' = 2a = \text{constant} \dots (3)$$

$$\text{at } t = 0, v = b = -ve$$

(v - t) graph is linear by equation (2)

with +ve slope

so ans is (3)

$$32)$$

$$v = 4t^3 - 2t$$

$$\int v dt = x = t^4 - t^2 \text{ and } x = 2 \text{ at } t = \sqrt{2} \text{ s}$$

$$a = 12t^2 - 2$$

$$a = 12(\sqrt{2})^2 - 2 = 22$$

$$33) \frac{p}{q} = \frac{x}{x+3x} = \frac{1}{4}$$

$$\Rightarrow p + q = 5$$

34)

$$v = v_0 - at$$

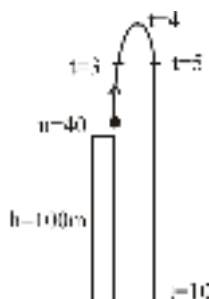
$$KE \propto v^2$$

$$\propto (v_0 - at)^2$$

Hence KE vs t is parabolic.

$$\text{Also } \frac{d(KE)}{dx} = \frac{d\left(\frac{1}{2}mv^2\right)}{dx} \propto \frac{v dv}{dx}$$

$\propto a$ which is (-ve) constant



35)

36)

displacement = Area under curve v-t

$$\frac{1}{2} (4) (10) + 10(\Delta t) = 120$$

$$\Delta t = 10s$$

$$\text{so total time} = 4 + \Delta t$$

$$= 14s$$

37) Theory

$$38) \frac{\alpha \beta t^2}{2(\alpha + \beta)} = 3 \text{ km}$$

$$\frac{10 \times 20t^2}{2 \times 30} = 3000$$

$$t = 30 \text{ sec}$$

$$V_{\text{avg}} = \frac{3000}{30} = 100 \text{ m/s}$$



39)

$$T_{ABCD} = 6s \text{ and } T_{CD} = 2s$$

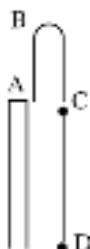
$$\Rightarrow T_{ABC} = 4s$$

$$\Rightarrow \frac{2u}{g} = 4$$

$$\Rightarrow u = 20 \text{ m/s}$$

$$C \text{ to } D \Rightarrow s = ut + \frac{1}{2} at^2$$

$$= 20(2) + \frac{1}{2} (10) (2)^2 = 60 \text{ m}$$



40) $T_{ABCD} = 6s \text{ and } T_{CD} = 2s$

$$\Rightarrow T_{ABC} = 4s$$

$$\Rightarrow \frac{2u}{g} = 4$$

$$\Rightarrow u = 20 \text{ m/s}$$

$$C \text{ to } D \Rightarrow s = ut + \frac{1}{2} at^2$$

$$= 20(2) + \frac{1}{2} (10) (2)^2 = 60 \text{ m}$$

41) Body will speed up when sign of v and a is same. So at Q and R it is speeding up.

42) $v = 8 - 2t$

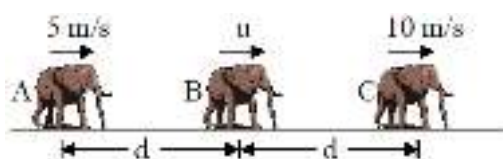
$$v = 0 \text{ at } t = 4s$$

so particle changes direction at $t = 4s$.

$$\text{distance} = |x(t = 4) - x(t = 0)| + |x(t = 6) - x(t = 4)|$$

$$|[8(4) - 4^2 + 2] - [0 - 0 + 2]| + |8(6) - 6^2 + 2 - (8 \times 4 - 4^2 + 2)|$$

$$= 20 \text{ m}$$



43)

$$(u - 10)t = d, \dots (1)$$

(t is time taken to catch C by B)

When B catches C separation between A & C becomes $3d$, initially it is $2d$.

Relative displacement $d = (10 - 5)t$... (2) by equation (1) & (2)

$$u - 10 = 10 - 5$$

$$u = 15 \text{ m/s.}$$

44)

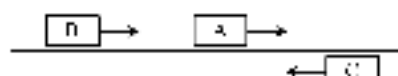
Coin is dropped from lift so their initial relative velocity is zero.

$$S_{\text{rel}} = 0 \times t + \frac{1}{2}gt^2 = 5$$

time $t = 1 \text{ s}$ to reach the floor

$$\text{total distance travelled by coin } S_{\text{total}} = 10t + \frac{1}{2}gt^2$$

$$\text{put } t = 1, S_{\text{total}} = 10 + 5 = 15 \text{ m}$$

45)  $u_A = 36 \times \frac{5}{18} = 10 \text{ m/s,}$

$$u_B = 54 \times \frac{5}{18} = 15 \text{ m/s}$$

$$u_C = 54 \times \frac{5}{18} = 15 \text{ m/s}$$

Initially $AB = AC = 1 \text{ km} = 1000 \text{ m.}$

To safely overtake A by B,

Let acceleration of B is a .

$$AB = (15 - 10)t + \frac{1}{2}at^2 \quad \dots(1)$$

$$AC = (15 + 10)t = 1000 \quad \dots(2)$$

$t = 40 \text{ s}$ put in (1)

$$5 \times 40 + \frac{a}{2} \times (40)^2 = 1000$$

$$a \times 800 = 1000 - 200$$

$$a = 1 \text{ m/s}^2$$

CHEMISTRY

46) $t_{1/2} \propto \frac{1}{(A_0)^{n-1}}$

OR

$$t_{1/2} \propto \frac{1}{(A)_0}$$

$$20 \propto \frac{1}{1 \times 10^{-2}} \quad \dots\dots(1)$$

$$t_{1/2} \propto \frac{1}{2 \times 10^{-2}} \quad \dots\dots(2)$$

$$\frac{(2)}{(1)} \quad \frac{t_{1/2}}{20} = \frac{\frac{1}{2 \times 10^{-2}}}{\frac{1}{1 \times 10^{-2}}} = \frac{1}{2}$$

$$t_{1/2} = 10 \text{ min}$$

$$47) \pm \frac{\Delta c}{\Delta t} = \pm \frac{\Delta P}{\Delta t} \times \frac{1}{RT}$$

48)

$$\log \frac{k}{A} = -\frac{E_a}{2.303RT}$$

$$\log 3.8 \times 10^{-18} = -\frac{E_a}{2.303 \times 8.314 \times 300}$$

$$E = 99.982 \text{ KJ/mol}$$

$$\simeq 100 \text{ KJ/mol}$$

$$49) C_t = C_0 e^{-kt}$$

$$= 10 e^{-k \times \frac{1}{k}}$$

$$C_t = 10 e^{-1}$$

$$C_t = \frac{10}{e}$$

$$\text{rate} = K[C_t] = 500 \times \frac{10}{e}$$

$$\text{Rate} = \frac{5000}{e}$$

$$50) \therefore K = \frac{2.303}{t} \log \left(\frac{V_\infty}{V_\infty - V_t} \right)$$

V_∞ = vol. of gas collected at the end of reaction

V_t = vol. of gas collected at any time 't'

$$\text{at } t = 20 \text{ min.}$$

$$K = \frac{2.303}{20} \log \left(\frac{10}{10-5} \right) = \frac{2.303}{20} \log 2 \quad \dots(1)$$

$$\text{at } t = 10 \text{ min.}$$

$$K = \frac{2.303}{40} \log_{10} \left(\frac{10}{10-V_t} \right) \quad \dots(2)$$

$$\text{from (1) \& (2)}$$

$$\frac{2.303}{20} \log 2 = \frac{2.303}{40} \log \left(\frac{10}{10-V_t} \right)$$

$$2 \log 2 = \log \left(\frac{10}{10-V_t} \right)$$

$$\log 2^2 = \log \left(\frac{10}{10-V_t} \right) \Rightarrow \frac{10}{10-V_t} = 4$$

$$V_t = 7.5 \text{ mL}$$

51) All collisions of reactant molecules do not lead to product formation.

$$\begin{aligned}
 52) \quad 36 \text{ g O}_2 &= \frac{36}{32} = 1.125 \text{ mol L}^{-1} \text{ min}^{-1} \\
 -2 \frac{d[\text{H}_2\text{O}_2]}{dt} &= \frac{d[\text{O}_2]}{dt} \\
 \frac{d[\text{H}_2\text{O}_2]}{dt} &= 2 \times \frac{d[\text{O}_2]}{dt} = 2 \times 1.125 \\
 &= 2.25 \text{ mol L}^{-1} \text{ min}^{-1}
 \end{aligned}$$

53) A. Question Explanation

We are asked to find how the **half-life** ($t_{1/2}$) of a reaction of **order** n depends on the **initial concentration** a of the reactant.

B. Given Data

- A. Reaction order: n
- B. Initial concentration: a
- C. Concept: Half-life of **n^{th} -order** reaction

C. Concept

The formula for half-life $t_{1/2}$ of an **n^{th} -order reaction** ($n \neq 1$) is:

$$t_{1/2} \propto \frac{1}{a^{n-1}} \Rightarrow t_{1/2} \propto a^{1-n}$$

For a **first-order reaction**, $t_{1/2}$ is constant (independent of a).

For other orders, the half-life depends on the initial concentration.

D. Mathematical Calculation

$$t_{1/2} \propto a^{1-n}$$

This directly matches **Option 3**.

E. Final Answer

Option 3

54)

$$\text{Rate law} = r = k[\text{A}]^1[\text{B}]^1$$

its a second order reaction so unit of rate constant will be $\text{L mol}^{-1} \text{sec}^{-1}$

$$55) K = A.e^{-E_a/RT}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

56) NCERT XII, Part-I, Pg # 111

$$t_{87.5\%} = 3t_{1/2} \text{ (for first order reaction)}$$

57)

$$\Delta T_f \propto i$$

$$T_f \propto \frac{1}{i}$$

58) for ideal solution, $\Delta G_{\text{mix}} < 0$

$$\frac{P_A^0 - P_S}{P_A^0} = X_B$$

$$\Rightarrow \frac{10}{P_A^0} = 0.2$$

$$P_A^0 = 50$$

$$\frac{P_A^0 - P_S}{P_A^0} = X_B$$

$$\Rightarrow \frac{20}{50} = X_B$$

$$X_B = 0.4 \quad X_A = 0.6$$

60)

$$\text{Depression in F.Pt} \propto \frac{1}{F.pt.} \propto iC$$

61)

$$\Delta T_f = i K_f m$$

$$i \text{ for HX} = 1 + \alpha = 1 + 0.3 = 1.3$$

$$62) \Delta T_f = K_f \times m$$

$$= K_f \times \frac{w_2}{M_2 \times W_1} \times 1000$$

$$\frac{\Delta T_f}{K_f} = \frac{w_2}{M_2 \times W_1} \times 1000$$

(1 litre water = 1000 grams)

$$w_2 = \frac{\Delta T_f}{K_f} \times \frac{M_2 \times W_1}{1000}$$

$$= \frac{1}{\frac{1000}{180}} \times \frac{180 \times 1000}{1000}$$

$$= \frac{180}{1000} = 0.18 \text{ gram}$$

63)

With the same molality, i and K_b (depends on solvent only) - the two solutions have same ΔT_b

64)

$$\pi = \pi_1 + \pi_2$$

$$\begin{aligned}
 &= C_1RT_1 + C_2RT_2 \\
 &= \left(\frac{0.6 \times 3}{60 \times 0.6} + \frac{3.42 \times 3}{342 \times 0.6} \right) RT = \left(\frac{1}{6} + \frac{1}{20} \right) RT \\
 &= \left(\frac{1}{10} \right) \times 0.082 \times 293 \\
 P &= 2.4 \text{ atm}
 \end{aligned}$$

65)

Generated by Allie

Problem Statement: The question asks you to judge the correctness of two statements about azeotropes: the assertion claims that azeotropes are binary mixtures formed by ideal solutions, and the reason states that azeotropes boil without any change in composition. You need to decide which statements are correct and whether the reason correctly explains the assertion.

Underlying Concept: An azeotrope is a mixture of two or more liquids that boils at a constant temperature and whose vapor has the same composition as the liquid mixture, meaning the composition does not change during boiling. Importantly, azeotropes occur due to non-ideal behavior in mixtures, which means the solution shows deviations from Raoult's law (either positive or negative deviations). Ideal solutions follow Raoult's law exactly and do not form azeotropes.

Tips and Tricks: Recall that azeotropes are special mixtures with constant boiling points and composition; these cannot occur in ideal solutions that strictly follow Raoult's law.

Common Mistakes: Many students incorrectly think azeotropes form from ideal solutions because of the constant boiling point feature, but this property actually stems from non-ideal interactions.

Why Other Options Are Incorrect?: If someone selects that azeotropes form from ideal solutions, this is incorrect because ideal solutions have no deviation from Raoult's law and thus cannot form azeotropes. Also, confusing constant boiling composition with ideality leads to misunderstanding.

66) 1. Explaining Question

The question asks to determine which statement is true regarding the reaction and the changes in oxidation states of bromine and carbonate.

2. Concept

- A. Oxidation is the process where an element loses electrons and its oxidation state increases.
- B. Reduction is the process where an element gains electrons and its oxidation state decreases.
- C. To determine whether bromine and carbonate are oxidized or reduced, we need to assign oxidation numbers to these elements before and after the reaction.

3. Solution

- In the reaction: $3\text{Br}_2 + 6\text{CO}_3^{2+} + 3\text{H}_2\text{O} \rightarrow 5\text{Br}^- + \text{BrO}_3^- + 6\text{HCO}_3^-$

- A. Bromine in Br_2 has an oxidation number of 0. In Br^- , it has an oxidation number of -1, and in BrO_3^- , it has an oxidation number of +5.
- B. This means bromine is both reduced (from 0 to -1) and oxidized (from 0 to +5).
- C. The carbonate (CO_3^{2-}) has an oxidation number of +4 in CO_3^{2-} and remains at +4 in HCO_3^- , so carbonate is neither oxidized nor reduced.

4. Final Answer: Option 4. Bromine is both reduced and oxidized.

67) **Solution:**

(A)

- 3

NH_3

$$= x + 3 = 0$$

$$x = -3$$

(B)

+1

O_2F_2

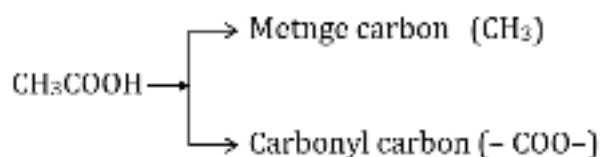
$$\Rightarrow 2x + 2(-1) = 0$$

$$\Rightarrow x = +1$$

(C)

+ 3

CH_3COOH ($\text{C}_2\text{H}_4\text{O}_2$)



We have to calculate for Carbonyl carbon for COO^-

$$\Rightarrow x + 2(-2) = -1$$

$$x = +3$$

(D)

BaO_2

$$\Rightarrow +2 + 2x = 0$$

$$(x = -1)$$

Option (3) is correct answer

68) **1. Explaining Question**

The question asks to determine the oxidation states of the two chlorine atoms in calcium hypochlorite (CaOCl_2).

2. Concept

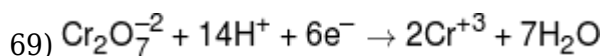
In peroxides, the oxidation number of oxygen is typically -1. In CaOCl_2 , the oxidation state of chlorine in

the Cl^- ion is -1, and in the OCl^- ion, chlorine has an oxidation state of +1.

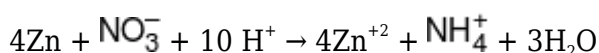
3. Solution

- In CaOCl_2 , the oxidation state of Cl^- in Cl^- is -1, and in OCl^- , it is +1.

4. Final Answer (1) -1 and +1



70)



71) 1. Explaining Question

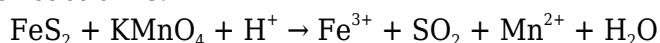
The question involves calculating the equivalent mass of FeS_2 in a redox reaction.

2. Concept

Equivalent mass is the molar mass divided by the n-factor, where the n-factor represents the total change in oxidation state of the atoms involved.

3. Solution

The reaction is:



The oxidation state of Fe in FeS_2 is +2, and it changes to +3 in Fe^{3+} (a change of 1).

The oxidation state of S in FeS_2 is -1, and it changes to +4 in SO_2 (a change of 5 per sulfur atom, with 2 sulfur atoms, so the total change is 10).

The total change in oxidation state for FeS_2 is $1 + 10 = 11$.

4. Final Answer: Option 1. $M/11$ (where M is the molar mass of FeS_2)

72)

In NH_4NO_3 ; OS + N is
-3 & +5

73)

Concept based formula

74)

Data base order

75)

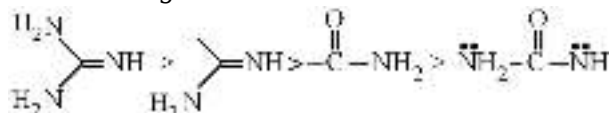
XIIth NCERT

76)

Stability of C^+ $\propto$ Resonance
 $\propto +M > +H > +I$

77)

Basic strength order



78)

More resonance $\rightarrow$ More stable C^+

79)

At C_1 & C_5 SIR effect is observed.

80)

Position of atom never change

81)

Follow Hukel rule

82)

More basic easily protonated

83) Cyclic resonance must be present b/w 2π , 6π , 10π , 14π , ..., e^- .

84)

XIIth NCERT

85) Stability of carbocation is given on the basis of more resonance.

86)

$+M \rightarrow$ -ve charge / lone pair on link atom

87)

More resonance $\rightarrow$ More stable C^+

88)

Data base

89)

α carbon sp^3 hybrid with minimum one α_H

90)

Data base

BIOLOGY

91)

Old NCERT, Pg. # 115

92)

Old NCERT, Pg. # 113

93)

Old NCERT, Pg. # 114

94)

Old NCERT, Pg. # 112

95)

Old NCERT, Pg. # 111

96)

Old NCERT, Pg. # 112

97) NCERT (XI) Pg. # 114

98)

NCERT (XIth) Pg. # 113

99) NCERT XIth P.No. 113

100) Pg. No. 116 NCERT 2022 - 2023 Edition

101) **Diaphragm:** Frogs do not have a diaphragm for respiration. They breathe through their skin and lungs.

Anus: Frogs have a cloaca, a single opening for the digestive, urinary, and reproductive systems.

Tail: Frogs do not have tails in their adult stage. Cortex, medulla and pyramid absent in kidney of frog ans-1

102)

New NCERT, Pg. # 83

103)

New NCERT, Pg. # 80

104)

New NCERT, Pg. # 83

105)

New NCERT, Pg. # 82

106)

NCERT Pg.#118

107)

Old NCERT, Pg. # 101

108)

Old NCERT, Pg. # 102

109)

Old NCERT, Pg. # 101

110)

Old NCERT, Pg. # 103

111)

Old NCERT, Pg. # 104

112)

Old NCERT, Pg. # 101

113)

Old NCERT, Pg. # 102, 103

114)

NCERT-XI, Pg. No. # 44

115)

The correct answer is **3 Nereis**.

- A. **Dioecious** means that a species has separate male and female individuals.
- B. **Nereis** (a type of marine worm) are dioecious, meaning they have distinct male and female individuals.

116)

The correct answer is Option 3: Echinodermata

- A. Echinoderms exhibit radial symmetry as adults. This means their bodies can be divided into similar halves along multiple planes passing through the central axis. Examples include starfish and sea urchins.
- B. However, the larvae of echinoderms have bilateral symmetry. This indicates their evolutionary relationship to other bilaterally symmetrical animals.

117)

NCERT-XI, Pg. No. # 42

118)

NCERT-XI, Pg. No. # 39

119)

The correct answer is Option 4: Ichthyophis

- A. Ichthyophis belongs to the order Gymnophiona, which are commonly known as caecilians.
- B. Caecilians are limbless amphibians, meaning they lack legs.

120)

NCERT-XI, Pg. No. # 46, 47

121)

NCERT-XI Pg. # 47, 48

122)

NCERT Pg. # 41

123)

NCERT-XI, Pg. No. # 38

124)

NCERT-XI, Pg. No. # 49, 50

125)

NCERT-XI, Pg. No. # 44

126)

NCERT XI, Pg # 53 to 55

127) NCERT, Pg # 46, 48

128)

NCERT-XI, Pg. No. # 47

129) NCERT, Pg # 44

130) NCERT Pg. # 54

131)

NCERT-XI, Pg. No. # 41, 45, 48

132) NCERT, Pg.# 44

133) NCERT Pg # 54

134)

NCERT-XI, Pg. No. # 46

135) NCERT Pg. # 51

136) NCERT-XIth, Pg. # 13, 24, 29, 32

137) NCERT-XIth, Pg. # 26

138) NCERT-XIth, Pg. # 26,30,32, 33

139) NCERT-XIth, Pg. # 29, 33

140) NCERT-XIth, Pg. # 33

141) NCERT-XIth, Pg. # 29, 32

142) NCERT-XIth, Pg. # 30, 32

143) NCERT-XIth, Pg. # 29, 30

144) NCERT-XIth, Pg. # 30, 32

145) NCERT-XIth, Pg. # 24,27,33

146) NCERT-XIth, Pg. # Module 1

147) NCERT-XIth, Pg. # 26,27

148) NCERT-XIth, Pg. # 29

149) NCERT-XIth, Pg. # 29, 32

150) NCERT-XIth, Pg. # 32

151) NCERT-XIth, Pg. # 33

152) NCERT-XIth, Pg. # 29, 32, 33

153) NCERT-XIth, Pg. # 24, 26

154) NCERT-XIth, Pg. # 29, 30

155) NCERT (XII) ; Page No. # 32

156) NCERT-XIth, Pg. # 28

157) NCERT-XIth, Pg. # Module 1

158) NCERT-XIth, Pg. # 32

159) NEW NCERT-XI, Pg # 7

160)

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161)

NCERT (XI) Pg. # 4

162)

NEW NCERT XI pg no. 8

163) NCERT XI Pg # 2

164) NCERT Pg. # 7, 8

165) NCERT XI Pg. # 13

166) NCERT XI Pg # 11, 12

167)

NCERT-XI, Pg # 18

The number of fungi from the provided list that have septate and branched mycelium with the absence of asexual spores is 1, specifically *Agaricus*.

168) NCERT XI, Pg. # 15

169)

NCERT-XI, Pg # 14, 15

170)

NCERT-XI, Pg # 14

171)

NCERT-XI, Pg # 90, 91

172) NCERT-XI Pg # 13

173)

NCERT-XI, Page No. # 14

174) NCERT-XI, Pg. # 13

175)

NCERT-XI, Pg # 13,14,15

176) NCERT XI, Pg. # 14,15

177) NCERT-XI, Pg # 16

178) NCERT-XI, Pg # 17,18

179) NCERT (XI) Pg. # 17,18

